Combustion of Solids

16

STUDIES ON COMBUSTION OF CARBON PARTICLES IN FLAMES AND FLUIDIZED BEDS

By SAKAE YAGI AND DAIZO KUNII

General Theory

COMBUSTION OF SINGLE CARBON PARTICLE

Not only the chemical reaction rate coefficient of a carbon particle but the mass transfer coefficient through the boundary film surrounding it must be considered when studying the overall rate of combustion of a carbon particle, especially in the case of combustion in flames or in fluidized beds. The value of the chemical reaction rate coefficient k_{c1}[cm/s] is calculated, for example, by changing the unit of the specific combustion rate K_c in the following equation derived by Parker and Hottel[1].

$$10^3 K_c = (9.55 \times 10^9/\sqrt{T})P_1 \exp(-44{,}000/\mathbf{R}T) \quad (1)$$

The value of the mass transfer coefficient through the boundary film around the carbon particle, k_{f1}[cm/s], can be calculated by the following equation, derived from experimental data on mass transfer for various conditions by N. Froessling and by Ranz and Marshall[2].

$$\mathbf{N}_{Nu}{}' = k_f D_p \xi_f / \mathbf{D}_{vf} = 2.0 + 0.6(\mathbf{N}_{Sc})^{1/3}(\mathbf{N}_{ReM})^{1/2} \quad (2)$$

The combustion rate of a single carbon particle of radius r according to $C + O_2 \rightarrow CO_2$ can be described as follows:

$$-\frac{dr}{d\theta_p} = \alpha k_1 C_1 \qquad \frac{1}{k_1} = \frac{1}{k_{c1}} + \frac{1}{k_{f1}} \quad (3)$$

When a particle reacts with both oxygen and carbon dioxide during combustion, the subsequent equation can be given.

$$-\frac{dr}{d\theta_p} = \alpha \bar{z} \qquad \bar{z} = \bar{k}_1 \overline{C}_1 + \bar{k}_2 (\overline{C_2 - C_2^*}) \quad (4)$$

In another ideal case where the ash layer remains around the unburned core of the particle and prevents the diffusion of gas (Fig. 1), the equations given by the authors can be used to obtain the combustion rate at the core[3]:

$$-\frac{dr}{d\theta_p} \doteqdot \alpha \bar{k}_1 C_1 \qquad \frac{1}{\bar{k}_1} = \frac{1}{k_{c1}} + \frac{1}{k_{d1}} + \frac{1}{3k_{f1}} \qquad k_{d1} = \frac{12\mathbf{D}_{vs}}{D_p} \quad (5)$$

Generally speaking, in this case, where the chemical reaction, the diffusion of gas through the solid layer, and the mass transfer through the boundary film must be taken into consideration, the radius-ratio of unburned core r/R can be expressed by the following equations given by the authors[3, 4].

$$\frac{\theta_p}{\theta_B} = 1 - \frac{\gamma(r/R) + 3\omega\gamma(r/R)^2 + \omega(1 - 2\gamma)(r/R)^3}{\omega + \omega\gamma + \gamma} \qquad \omega = \frac{k_{c1}}{k_{d1}} \quad (6)$$

$$\theta_B = \frac{D_p}{2\alpha \bar{k}_1 C_1} = \text{time required for complete combustion of particle of diameter } D_p$$

The value of γ is also expressed as follows:

$$\gamma = \frac{3k_{f1}}{kd_1} = \left(\frac{\mathbf{N}_{Nu'}}{4\xi_f}\right)\left(\frac{\mathbf{D}_{vf}}{\mathbf{D}_{vs}}\right) \doteqdot \frac{\mathbf{N}_{Nu'}}{4e} \quad (7)$$

e = (total sectional area of small holes in the solid layer through which the gas can diffuse)/(unit area)

in case of boundary film controlling